$$I=\frac{V}{R}, I=\frac{240}{1.6}=150A$$

2) ***Three resistors 6Ω, 3Ω and 2Ω are connected in parallel. The current flow of this circuit is 2A. Find out the value of***
i) Total Resistance
ii) Voltage.

$R_1 = 6\Omega,\ R_2 = 3\Omega,\ R_3 = 2\Omega,\ I = 2A$
$R = ?\ \ V = ?$

When they are connected in parallel

Then
$$R=\frac{R_1R_2R_3}{R_2R_3+R_1R_3+R_1R_2}$$
$$=\frac{6\times3\times2}{(3\times2)+(6\times2)+(6\times3)}$$

$$R=\frac{36}{6+12+18}=\frac{36}{36}=1\Omega$$

According to ohm's law

$$I=\frac{V}{R}$$
$$2=\frac{V}{1}, V=2V$$

2.6.3 Series - Parallel Circuit

In series parallel circuit, one or more resistors are connected in series with more resistors connected in parallel. This is the combination of series parallel circuit.

Fig 2.7 represent, five resistors connected in series parallel circuit. Here R_1, R_2 are connected in series and R_3, R_4, and R_5 are connected in parallel. These parallelly connected resistors are connected in series with R_1 and R_2.

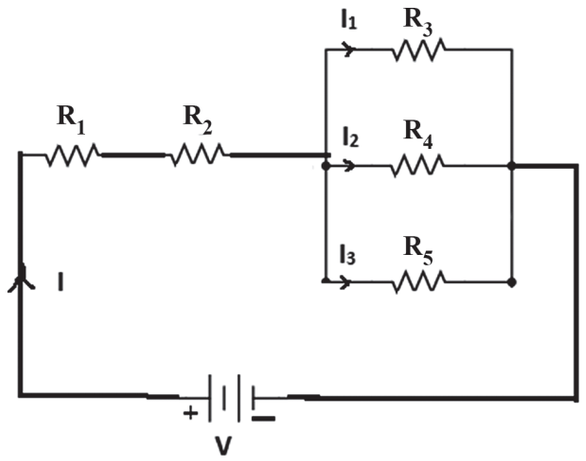


Fig 2.7 *Series - Parallel Circuit*

Hence the total resistance of the circuit is

$$R=R_1+R_2+\left[\frac{R_3\times R_4\times R_5}{R_4R_5+R_3R_5+R_3R_4}\right]$$

2.6.4 Kirchhoff's Laws

Scientist Kirchhoff derived two more laws based on ohm's law namely,

i) Kirchhoff's Current Law (KCL)
ii) Kirchhoff's Voltage Law (KVL)

Current law is based on the current and voltage law is based on the voltage. These laws are explained as follows.

Kirchhoff's Laws are used in determining the equivalent resistance of a complex network and the current flowing in the various conductors.

i) Kirchhoff's Current Law (KCL)

The sum of the current flowing towards a point (i.e junction) is equal to the sum of the current flowing away from the point. In other words, the algebraic sum of

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